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METHODOLOGIES AND APPLICATION

An adaptive hybrid evolutionary algorithm and its application in aeroengine maintenance scheduling problem

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Abstract
Multi-objective evolutionary algorithms (MOEAs) have been successfully employed to solve many scientific and engineering problems. However, many algorithms perform ill in maintaining diversity and convergence simultaneously. In this paper, we devised a novel operator selection framework based on two collaborative indicators, generational distance (GD) and maximum spread (MS) to improve the diversity while maintaining a good convergence. By calculating the variation of GDs and MSs over the past 7 iterations, an instruction is conveyed to select a proper operator to execute next 7 iterations. This process is repeated until it reaches the maximum iteration. Two operators are embedded in this algorithm which are differential evolution operator (DE/rand/1) and our proposed crow search operator which is deemed to be efficient in explorating the search space. MOEA/D is utilized as the basis framework of our proposed algorithm. Experiments indicate that our proposed algorithm is valid and outperforms other famous algorithms in terms of diversity and convergence. In the end, a particular aeroengine maintenance scheduling problem is solved by our proposed algorithm.

Keywords Multi-objective evolutionary algorithms · Collaborative indicator-based operator selection · Differential evolution · Crow search · Maintenance scheduling problem

1 Introduction

Aeroengine, the propulsion subsystem of an aircraft, generates the thrust essential to accomplish a flight. It is vital to maintain the high reliability of the aeroengine for an aircraft (Li et al. 2018a). Complex systems require not only reliability analysis (Li et al. 2019; Mi et al. 2018), but also maintainability analysis. Aircraft engines must operate reliably to accomplish the aircraft's specified mission. To achieve this goal, effective maintenance policy has to be implemented. Typically, the maintenance of aeroengines occurs in two occasions: unscheduled maintenance and scheduled/routine maintenance (Kleeman and Lamont 2005). Unscheduled maintenance refers to "fix it when it breaks" concept, which confines its application confronting the unpredictable uncertainty in repair jobs. It would be profound to renovate a component before it breaks. Scheduled maintenance has shown its superiority in this scenario and been applied in aeroengine maintenance (Avelino et al. 2016; Fornlöf et al. 2016b; Papakostas et al. 2010; Safaei et al. 2011). However, the prescribed scheduled maintenance strategy is incontrovertibly inclined to increase the maintenance cost compared with the unscheduled maintenance. It is vital to seek a balance between a prescribed performance and reasonable scheduled maintenance amount. This refers to a MOOP. Rare references are focusing on the trade-off issues involved in the scheduled maintenance of aeroengines, see (Fornlöf et al. 2016a; Kleeman and Lamont 2005, 2007; Kumar

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et al. 1999; Painter et al. 2006). However, the main contribution of the rare researches (Fornlöf et al. 2016a; Kleeman and Lamont 2005, 2007) puts more efforts on the formulation of the maintenance problem rather than the effectiveness of the optimization problems.

Multi-objective optimization is a typical trade-off problem involved in many practical applications, e.g., engineering design, economic allocation and operations research (Deb and Jain 2003; Geoffrey and Ilan 2008; Lee et al. 2008; Li et al. 2018b, 2015; Miettinen 1999; Zhou et al. 2011). Cognitively speaking, it is impossible to optimize all conflicting objectives simultaneously. An improvement in arbitrary objective will induce a decline in one or more of the others. Hence, the result of a MOOP is inevitably a set of trade-off solutions (Pareto Optimal solutions) rather than one single compatible solution. Most researchers, during the past decades, are devoted to obtain a closest approximation to the accurate Pareto optimal solutions (defined in the next section). In the early stage, the mathematics method, especially the scalarization method is prevalent, of which the basic idea is to transfer multiple objectives into one single objective by formulating a scalarizing/aggregating function (Ehrgott 2005; Miettinen and Mäkelä 2002). However, the scalarization methods are susceptible to the actual Pareto front (defined in the next section), they would probably end up with a sub-optimal solution (algorithm premature stagnation with respect to Pareto-based methods) when confronting with a concave or a disconnect front (Thakur et al. 2014). Different from the conventional mathematics method, multi-objective evolutionary algorithms (MOEAs), originating from the Darwinian Theory, generate a set of trade-off solutions rather than one single optimum solution in each iteration which promotes the efficiency and adaptability (Deb 2014; Deng et al. 2020a, c; von Lücken et al. 2014). Furthermore, MOEAs alleviate the computation cost by substituting the derivative calculation with the natural selection strategy.

As the research problem progresses, it is found that as the number of objectives increases (objective function greater than 3), the commonly used MOEAs based on the basic framework of Pareto domination, such as non-dominated sorting genetic algorithm (NSGA-II) (Deb et al. 2002) and improved strength Pareto evolutionary algorithm (SPEA2) (Zitzler et al. 2001) experience a dramatic performance degradation (Giagkiozis and Fleming 2015; Purshouse and Fleming 2003). In view of this, MOEAs based on relaxed dominance relations (Laumanns et al. 2002), MOEAs based on performance metrics (Bader and Zitzler 2011; Beume et al. 2007), MOEAs based on decomposition (Zhang and Li 2007), and MOEAs based on preference information (Wang et al. 2015) have emerged and demonstrated better performance than Pareto dominance-based MOEAs in dealing with MaOPs. However, there are still some shortcomings in these types of methods, and they are still in the exploration and improvement stage. How to better balance the diversity and convergence of optimization solutions of various types of MOEAs in the objective space is a current concern for many scholars. Wang et al. (2017) devised a hybrid MOEA with adaptive multi-population strategy (HMOEA-AMP) to take advantage of PSO in convergence to construct the external archive and DE in diversity to update external archive. Similarly, Maghawry et al. (2021) devised a hybrid strategy by combining GA and PSO, with the help of K-Means as a criterion to determine when to introduce PSO to rehabilitate the solutions rejected by GA to enhance the diversity. The basic idea of both of these papers is to balance the diversity and convergence in the objective space with the help of different heuristic algorithms, with little in-depth discussion addressing the exploration and exploitation of EAs in the search space. Therefore, in this paper, we focus on the balance between diversity and convergence of MOEAs while further exploring the impact of the balance between exploration and exploitation of MOEAs in the search space on the performance of MOEAs.

The remainder of this paper is organized as follows: Sect. 2 introduces the basic terminology and notation of MOOPs. Section 3 elaborates the principle of selection of different operators based on progressively generated indicators. In Sect. 4, experiments are conducted to compare the performance of the proposed algorithm with other elite algorithms. In Sect. 5, the proposed algorithm is utilized to optimize a job-based aeroengine maintenance scheduling problem. Section 6 enclosed this paper with a conclusion and further research on MOEAs.

2 Definition and basic concepts

In this paper, we address the MOOPs, in the form of minimization (Miettinen 1999),

$$\begin{aligned} &\min_{\mathbf{x}\in D} \mathbf{f}(\mathbf{x}) = \min\{f_1(\mathbf{x}), f_2(\mathbf{x}), \ldots, f_m(\mathbf{x})\}, \\ &\text{s.t.} \quad \mathbf{x} = (x_1, x_1, \ldots, x_n)^T \in D, \end{aligned} \tag{1}$$

where $\mathbf{f}(\mathbf{x})$ is the vector of objectives to be minimized, $\mathbf{x}$ is the vector of decision variables within the feasible domain of $D \subseteq \mathbb{R}^n$, m is the number of objective functions, n is the number of decision variables.

The term of Pareto optimality is the fundamental concept in MOOPs, of which the main definitions are listed below (Elsayed et al. 2014).

Definition 1: Pareto dominance A n-dimensional vector $\mathbf{u} = \{u_1, u_2, \ldots, u_n\}^T$ is said to dominate another vector

$\mathbf{v} = \{v_1, v_2, \ldots, v_n\}^T$ (denoted as $\mathbf{u} \prec \mathbf{v}$) if and only if $\forall i = \{1, 2, \ldots, m\}$, the objectives of the two solutions satisfy $f_i(\mathbf{u}) \leq f_i(\mathbf{v})$ and $\exists j = \{1, 2, \ldots, m\}, f_j(\mathbf{u}) < f_j(\mathbf{v})$.

Definition 2: Weakly Pareto optimal A solution $\mathbf{u} = \{u_1, u_2, \ldots, u_n\}^T$ is said to be weakly Pareto optimal if and only if no other solution $\mathbf{v} = \{v_1, v_2, \ldots, v_n\}^T$ satisfies that $\exists j = \{1, 2, \ldots, m\}, f_j(\mathbf{u}) < f_j(\mathbf{v})$.

Definition 3: Pareto optimal solution A solution $\mathbf{x} = \{x_1, x_2, \ldots, x_n\}^T$ is Pareto optimal if there is not any solution $\mathbf{y} = \{y_1, y_2, \ldots, y_n\}^T$ that dominates it.

Definition 4: *PS* All the Pareto optimal solutions are called *PS*.

Definition 5: *PF* The points which are the image of *PS* under the objective function mapping are named *PF* (also named Pareto front).

Definition 6: Ideal objective vector The vector comprised by the of infimum of $f_i(\mathbf{x})$ for every $i = \{1, 2, \ldots, m\}$ is called ideal objective vector and denoted by $\mathbf{z}^* = \{z_1^*, z_2^*, ..., z_m^*\}$.

3 Collaborative indicator-based operator selection

As the performance of SBX and PM operators in MOEA/D (Zhang and Li 2007) depends on the optimization problem itself. Much efforts have been made to alleviate this dilemma by substituting DE operator for the SBX operator (Li and Zhang 2009). However, this is still not enough confronting with more complicated problems. To address the above scenario, AOS strategies (Gonçalves et al. 2015; Li et al. 2014) have recently become popular in evolutionary multi-objective optimization, with the basic concept of adaptively deciding which operator to employ during the iterative process. The two main attributes of the AOS strategy are reward assignment and reward-based operator selection. Since the selection process of AOS strategy is based on a probabilistic approach, this strategy is essentially based on random selection and is computationally intensive. In view of this, we proposed another operator selection mechanism based on two collaborative indicators. Moreover, the operator pool consists of only DE operators (Das and Suganthan 2011; Deng et al. 2020b; Li et al. 2011; Song et al. 2021) and modified CSA operators, which is computationally efficient.

The aim of this section is to propose a novel algorithm named collaborative indicator-based operator selection for MOEA based on decomposition (MOEA/D-IHO) in which different operators are adaptively selected based on collaborative indicator values.

3.1 CSA

CSA (Askarzadeh 2016; Fu et al. 2018), is a newly devised heuristic single objective optimization algorithm showing advantages in addressing the exploration issue in search space (Askarzadeh 2016). The basic procedure of CSA is elaborated as below.

At each iteration $iter = 1, 2, \ldots, iter_{\max}$(time), a crow has a memory $\mathbf{m}_i^{iter}, i = 1, 2, \ldots, N$ of its own hidden location. The current position of each crow is expressed as $\mathbf{x}_i^{iter} = [x_{i,1}^{iter}, x_{i,2}^{iter}, \ldots, x_{i,d}^{iter}]$. The memory updates with respect to the best location founded till *iter*. In a particular iteration *iter*, crow j intends to visit its memory location, meanwhile crow i randomly chooses crow j to follow. Two conditions take place, which are:

Condition 1: Crow j does not notice that crow i is tracing, crow i moves to the vicinity of the memory place of crow j and the new position of crow i is formulated as

$$\mathbf{x}_i^{iter+1} = \mathbf{x}_i^{iter} + r_i \cdot fl_i^{iter}(\mathbf{m}_j^{iter} - \mathbf{x}_i^{iter}) \tag{2}$$

where r_i denotes a randomly generated number between 0 and 1, fl_i^{iter} denotes the flight length of crow i at iteration *iter*. The flight trajectory of crow i is illustrated in Fig. 1. To enhance the diversity, $\left(\mathbf{m}_j^{iter} - \mathbf{x}_i^{iter}\right)^T$ is multiplied by the flight length fl_i^{iter} and the random generated number r_i, which would locate the position $\mathbf{x}_i^{iter+1}$ at the vicinity of $\mathbf{m}_j^{iter}$. It can be seen from Fig. 1 that with $r_i \cdot fl_i^{iter} > 1$ it tends to be a global search (diversification) while with $r_i \cdot fl_i^{iter} < 1$, it tends to be a local search around the local best (intensification).

Condition 2: Crow j knows that crow i is following and crow j would change its path and move to a wrong place to prevent its hidden location $\mathbf{m}_j^{iter}$ in the space.

The new location of crow i after *iter* is formulated as

$$\mathbf{x}_i^{iter+1} = \begin{cases} \mathbf{x}_i^{iter} + r_i \cdot fl_i^{iter}(\mathbf{m}_j^{iter} - \mathbf{x}_i^{iter}) & r_i \geq AW_j^{iter} \\ \text{random location} & r_i < AW_j^{iter} \end{cases} \tag{3}$$

where AW_j^{iter} denotes the awareness probability of crow j at iteration *iter*. If the randomly generated r_i is greater than AW_j^{iter}, it indicates that crow j does not notice the trace, and crow i would fly to a random place.

3.2 Modified CSA operator

The flight length (*fl*) and the awareness probability (*AP*) determine the principle of the search in CSA. It is very scarce in evolutionary multi-objective optimization society

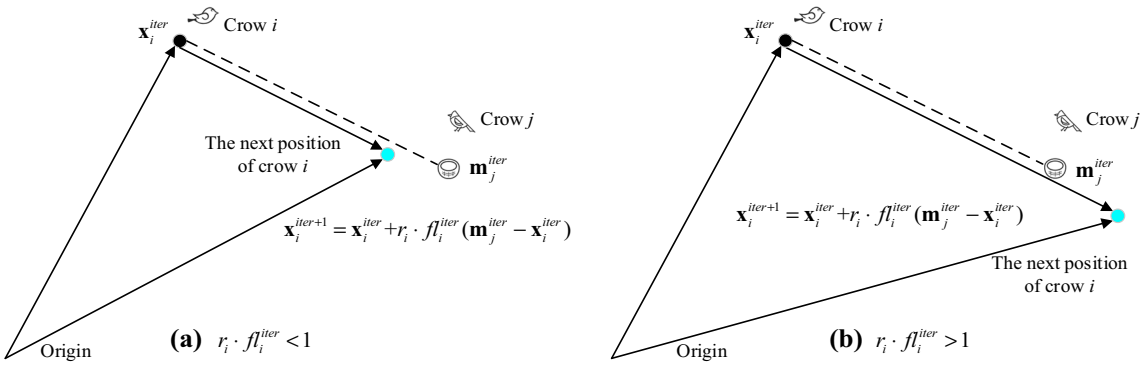


Fig. 1 The flight trajectory of CSA

to extend CSA to MOOP scenario by now (Fu et al. 2018; Hinojosa et al. 2018; Nobahari and Bighashdel 2017). All these applications mainly focused on the modification of the random search, the configuration of the awareness probability and the flight length in the original version.

Here, we extend the search mechanism to MOOP scenario by adding another crow k. As crow j in the original CSA tends to visit its memory place in the original version, this may result in local optimal. So the trajectory of crow j will be replaced by $(\mathbf{m}_k^{iter} - \mathbf{x}_j^{iter})$, and the tacking path of crow i will be

$$\mathbf{x}_i^{iter+1} = \mathbf{x}_i^{iter} + r_i \cdot fl_i^{iter}(\mathbf{m}_k^{iter} - \mathbf{x}_j^{iter} - \mathbf{x}_i^{iter})$$

. Further the random location in Eq. (3) is modified to be $\mathbf{x}_i^{iter} + r_i \cdot \|U - L\|$. The proposed CSO is then formulated as

$$\mathbf{x}_i^{iter+1} = \begin{cases} \mathbf{x}_i^{iter} + r_i \cdot fl_i^{iter}(\mathbf{m}_k^{iter} - \mathbf{x}_j^{iter} - \mathbf{x}_i^{iter}) & r_i \geq AW_j^{iter} \\ \mathbf{x}_i^{iter} + r_i \cdot \|U - L\| & r_i < AW_j^{iter} \end{cases} \tag{4}$$

where U and L are the upper bound and lower bound of the decision variables.

The CSO was examined to be exploration efficient in search space (Fu et al. 2018).

3.3 DE operator

The most popular mutation operation scheme is "DE/rand/1" (Das and Suganthan 2011) and formulated as

$$\mathbf{v}_i^{iter} = \mathbf{x}_{r_1^i}^{iter} + F(\mathbf{x}_{r_2^i}^{iter} - \mathbf{x}_{r_3^i}^{iter}) \tag{5}$$

where $\mathbf{v}_i^{iter}$ denotes the mutant vector with respect to the target vector $\mathbf{x}_i^{iter}$ at iteration *iter*, target vector, r_1^i, r_2^i, r_3^i are randomly generated integer within the range of [1, n], n is the number of the elements in a decision vector and different from i, F is a positive scaling factor.

The most popular crossover operation is binomial crossover and formulated as

$$u_{i,j}^{iter} = \begin{cases} v_{i,j}^{iter} & \text{if rand} \leq CR \text{ or } j = j_{rand} \\ \mathbf{x}_{i,j}^{iter} & \text{otherwise} \end{cases}, \quad j = \{1, 2, \ldots, n\} \tag{6}$$

where n is the number of the elements in a decision vector, CR is a another positive parameter, named crossover rate, which controls the percentage of the elements inherited from the mutant vector $v_{i,j}^{iter}$.

3.4 Collaborative indicator mechanism

MS (Adra and Fleming 2011) and *GD* (Van Veldhuizen and Lamont 1998) are two indicators that separately measure the maximum extent and the convergence, respectively. *C* (Zitzler et al. 2003) is utilized to compare two solutions in terms of convergence. During the iteration, algorithms sometimes stagnates into local optimal. We proposed a scheme to detect the stagnation and to instruct the operator utilized in the next iterations. The pseudo-code of MOEA/D-IHO is presented in Algorithm 1.

Algorithm 1: MOEA/D-IHO

Set: The population size N (identical to the size of weight vector set W), the neighbor size T, and the maximum iterations $iter_{\max}$.

1. Initialize the population $P^0=\{p_1^0,p_2^0,\ldots,p_N^0\}$ and a set of weight vectors $W=\{\mathbf{w}_1,\mathbf{w}_2,\ldots,\mathbf{w}_N\}$;
2. Calculate the ideal point z_0^* in accordance with P^0;
3. Determine the neighbors $\lambda_i^0=\{\mathbf{w}_i,\mathbf{w}_j,\ldots,\mathbf{w}_T\}$ of each weight vector $\mathbf{w}_i$;
4. Randomly assign each weight vector (subproblem) with a solution x_i^0 in P^0;
5. $iter\leftarrow 1$;
6. **while** $iter<iter_{\max}$ **do**
7. **if** $iter<7$
8. Execute function ***GenerateOffspring***(P^{iter},*DE*) to update the population P^{iter}
9. **elseif** mod($iter$,7)==0
10. Set judge parameter of C, $cj=0.1*(1-0.9*iter/iter_{\max})$, set judge parameter of GD, $gdj=0.01*(1-0.9*iter/iter_{\max})$;
11. Execute function ***CalIndicator***($\{P^{iter-1},\ldots,P^{iter-6}\}$) to generate a set of C, GD, and MS;
12. **if** $mean\{GD(i)\}<gdj$ or $mean\{C_diff\}>cj$
13. Execute function ***GenerateOffspring***(P^{iter}, DE) to update the population P^{iter}
14. **elseif** $std\{GD(i)\}>1e-4$
15. Execute function ***GenerateOffspring***(P^{iter}, CSO) to update the population P^{iter}
16. **else**
17. Execute function ***GenerateOffspring***(P^{iter}, (CSO, DE)) to update the population P^{iter}
18. end **if**
19. end **if**
22. end **while**
23. return P

Function 1: *GenerateOffspring*

Input: Operator type *OP*(DE/CSO), population at iteration $iter$, P^{iter}

Output: P^{iter}

1. **for** each subproblem $p_i^{iter}, i=1,2,\ldots,N$ (identified by weight vector) **do**
2. Generate an offspring u_i^{iter} from its neighbors by *OP* operator;
3. Execute function ***Update***(u_i^{iter}, p_i^{iter}) to update the subproblem p_i^{iter};
4. **end for**

Function 2: *Update*

Input: Mutant solution, u_i^{iter}, subproblem, p_i^{iter},

Output: Subproblem, p_i^{iter}

1. Calculate the Tchebyshev scalarizing function with respect to offspring u_i^{iter}, $f_{Tch}(u_i^{iter})$, and that of other neighbours' solutions, $f_{Tch}(nox_{i,k}^{iter})$, $nox_{i,k}^{iter}\in p_i^{iter}$, $,k=\{1,2,\ldots,T\}$
2. **for** each $nox_{i,k}^{iter}$ in the subproblem p_i^{iter} **do**
3. **if** $f_{Tch}(u_i^{iter})<f_{Tch}(nox_i^{iter})$ **then**
4. $nox_i^{iter}\leftarrow u_i^{iter}$
5. **end if**
6. **end for**

Function 3: *CalIndicator*

Input: Last 6 generations, $\{P^{iter-1},\ldots,P^{iter-6}\}$.

Output: A set of $\{C_diff(i),GD(i),MS(i)\},i=1,\ldots,5$.

1. $i\leftarrow 1$
2. **while** $i<=5$ **do**
3. Compute the C values between P^{iter-i} and $P^{iter-i-1}$, $C(P^{iter-i},P^{iter-i-1})$ and $C(P^{iter-i-1},P^{iter-i})$, and calculate the absolute difference value between these two MS values, $C_diff(i)=\left|C(P^{iter-i-1},P^{iter-i})-C(P^{iter-i},P^{iter-i-1})\right|$;
4. Compute the GD value between P^{iter-i} and $P^{iter-i-1}$, $GD(i)=GD(P^{iter-i},P^{iter-i-1})$;
5. Compute the MS value of P^{iter-i}, $MS(i)=MS(P^{iter-i})$;
6. **end while**

4 Experiment

4.1 Experiment settings

We selected the WFG2-WFG9 test suites (Huband et al. 2006), of which the decision variables are configured in accordance to (Wang et al. 2013) to conduct the experiment. In view of the goal of this paper, the MOEAs are tested in 2 objective spaces of these problems. The basic attributes of the test suites are listed in Table 1 (Huband et al. 2006).

In the experiment, 2 performance metrics are selected, which are the *IGD* (Czyżak and Jaszkiewicz 1998) and the *HV* (Zitzler and Thiele 1998). Both performance metrics can evaluate the diversity and convergence simultaneously. The larger *HV* value while the smaller *IGD* value, the better the MOEA's performance will be. Moreover, the median attainment surface (Fonseca and Fleming 1996) is introduced to intuitively display the obtained solutions.

In this chapter, 3 popular and efficient MOEAs, NSGA-II (Deb et al. 2002), MOEA/D (Zhang and Li 2007) and HypE (Bader and Zitzler 2011) are introduced to compare with our proposed MOEA/D-IHO. The population sizes of each peer algorithm are set to be identical to 100. The number of the weight vectors in the neighborhood of each weight vector T in MOEA/D is set to be 10 (Zhang and Li 2007). The number of sample to calculate the hypervolume in HypE is set to be 10,000 (Bader and Zitzler 2011).

The WFG parameters k and l are set to be 18 and 14. To evaluate the effectiveness of our proposed operator selection principle, the DE/rand/1 operator (Li and Zhang 2009; Li et al. 2011) and polynomial mutation (Deb et al. 2002) are used in all of these algorithms. The setting of all parameters in these algorithms remains the same. The distribution index η_m and mutation rate p_m in the polynomial mutation are set to be 20 and $1/n$, respectively, where n is the number of decision variables. The crossover rate CR and the scale factor in the DE/rand/1 are set to be 20 and 0.5.

31 runs of each MOEAs are performed to account for the discrepancy originating from the stochastic mechanism of EAs. 250 iterations are embedded in each run to alleviate possible time-consuming computing while ensuring an representative result.

4.2 Experiment results

Note that WFG*n*-*m* correspond to the WFG*n* with *m* objectives. It can be seen from Fig. 2 that,

(1) In face of the discrete convex WFG2 problem, all algorithms fail to converge to the true Pareto optimal frontier surface. Among them, MOEA/D-IHO

Table 1 Attributes of WFG test suite

Test functions	Separability	Modality	Bias	Geometry
WFG2	Nonseparable	Unimodal	Polynomial, flat	Convex, mixed
	Nonseparable	Multimodal	Polynomial, flat	
WFG3	Nonseparable	Unimodal	Polynomial, flat	Linear, degenerate
WFG4	Separable	Multimodal	Polynomial, flat	Concave
WFG5	Separable	Deceptive	Polynomial, flat	Concave
WFG6	Nonseparable	Unimodal	Polynomial, flat	Concave
WFG7	Separable	Unimodal	Parameter dependent	Concave
WFG8	Nonseparable	Unimodal	Parameter dependent	Concave
WFG9	Nonseparable	Multimodal, deceptive	Parameter dependent	Concave

outperforms the other algorithms in terms of convergence and diversity.

(2) In face of the linear convex WFG3 problem, all algorithms have good convergence, while MOEA/D-IHO outperforms the other algorithms in terms of diversity.

(3) In face of the multimodal separable concave WFG4 problem and the deceptive separable concave WFG5 problem, MOEA/D-IHO outperforms the other algorithms.

(4) In face of the nonseparable unimodal concave WFG6 problem, MOEA/D-IHO outperforms the other algorithms. Similarly, the combined performance of MOEA/D-IHO is significantly better than other algorithms in solving the WFG7 to WFG8 problems.

(5) In face of the nonseparable, multimodal, and deceptive WFG 9 problem, MOEA/D-IHO performs better in terms of diversity, while MOEA/D-DE performs better in terms of convergency.

Based on intuitive observations, MOEA/D-IHO outperforms the other algorithms on almost all tested functions. Due to the close distance between the optimized solution sets of each algorithm, it is also necessary to characterize the performance of each algorithm in terms of convergence synthetically with the help of box plots. It can be seen from Fig. 3 that our proposed MOEA/D-IHO performs best in almost all cases in terms of *HV*. It can be seen from Fig. 4 that our proposed MOEA/D-IHO also performs best in almost all cases in terms of *IGD* except for WFG6.

The results indicate that our proposed MOEA/D-IHO outperform all these famous algorithms with respect to complicated test problems in terms of *HV* and *IGD* metrics. In view of these experiment results, we are inclined to extend our proposed algorithm to solve practical 2 objective optimization problems.

5 Case study: aeroengine maintenance scheduling problem

5.1 Problem configuration

In this section, a job-based aeroengine maintenance scheduling problem (Kleeman and Lamont 2005, 2007) is introduced to demonstrate the application and performance of the proposed MOEA/D-IHO. In this particular job shop, n aeroengines are scheduled for maintenance, each broken engine is first disassembled into 5 components (fan, core, turbine, augmenter, and nozzle) to repair, then reassembled by 5 good components and tested to ensure functionality. It is assumed that each component corresponds to a specific maintenance area in the shop.

After all maintenances of these components are implemented, components in a particular aeroengine $e_i, i \in \{1, 2, \ldots, n\}$ are compared to distinguish the largest disparate one in accordance with the MTBF. If the disparity is large, the particular component $c_m^i, m \in \{1, 2, 3, 4, 5\}$ is swapped by another one c_m^j assembled in another engine $e_j, j \in \{1, 2, \ldots, n\}, j \neq i$ so that MTBFs of all components in this particular aeroengine e_i are similar in order to ensure e_i to preserve more time on wing and less time in maintenance.

5.2 Formulation of the maintenance scheduling problem

The competing objectives of the problem are to reducing the time consumed in the shop for repairs of all engines and matching the MTBFs of each component of all engines which are represented by makespan and aggregate swap count, respectively. The makespan corresponds to the test completion time of the last engine and is deemed to be as minimum as possible. The aggregate swap count corresponds to the matching MTBFs and is deemed to be as maximum as possible. These 2 objectives are competing with each other, as the first one tries to finish repair as

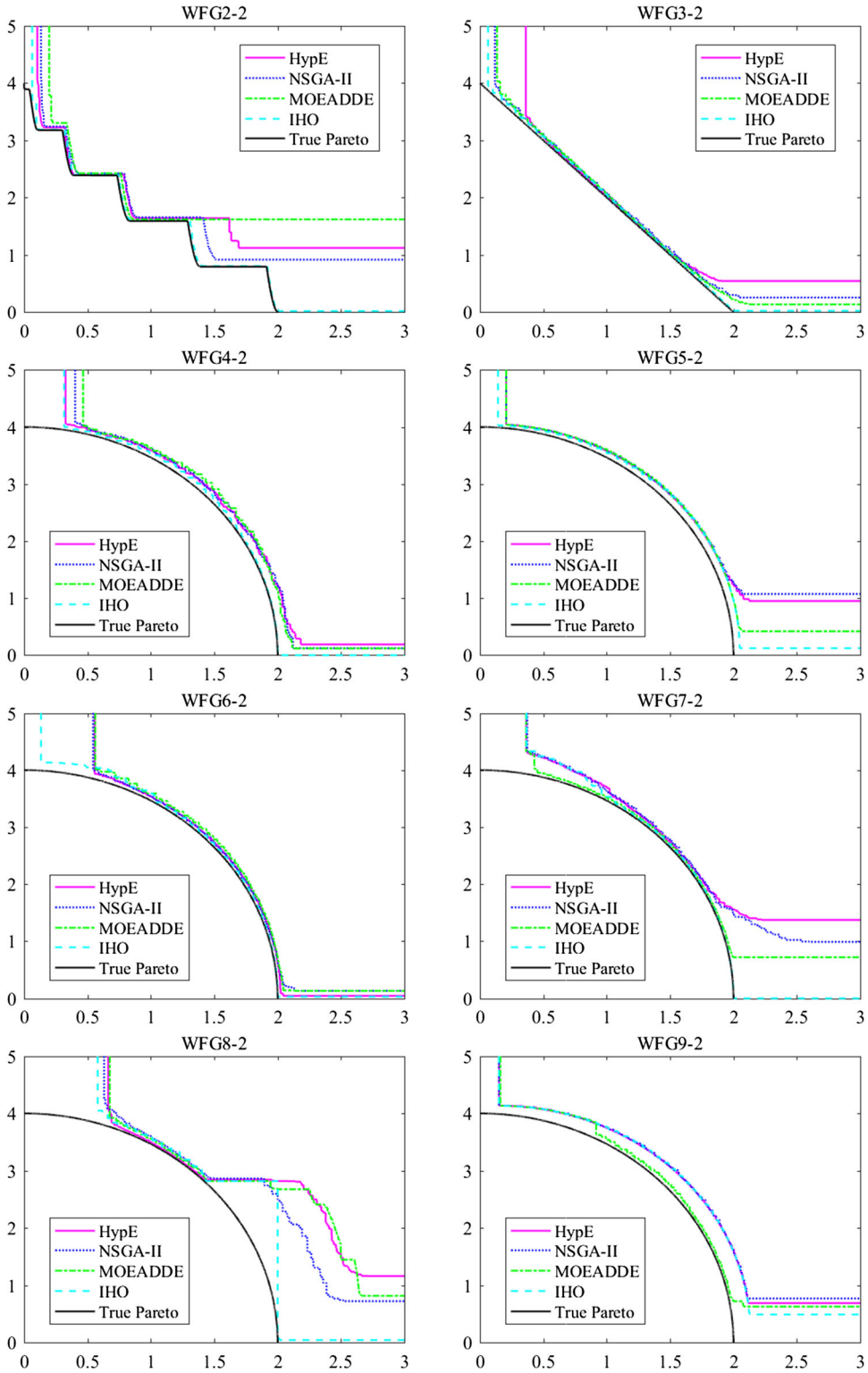


Fig. 2 Median attainment surfaces of 4 peer MOEAs

quickly as possible while the second one tries to swap as much as possible which slows down the repair time.

Prior to the formulation of the makespan (test completion time), engine repair time and possible swap are to be formulated.

The repair time for each engine is formulated as

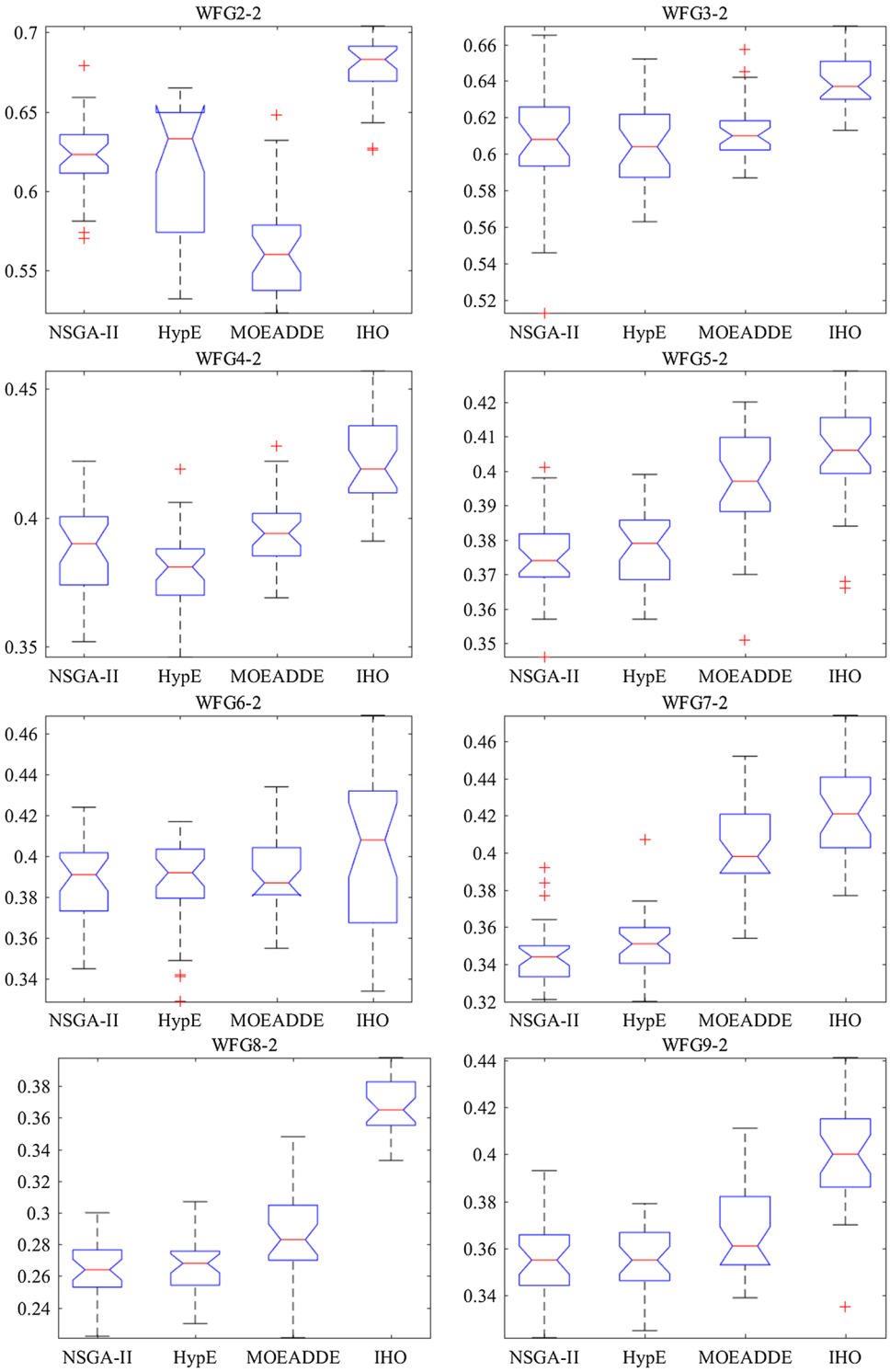


Fig. 3 Box plots of *HV* results

$$\forall e \in E, R_e = \max_{c \in C} t_{compStart} \sum_{i=1}^{P_e} t_R(i) P_e \tag{7}$$

where E is the set of engines, e is a particular engine, R_e is the repair time for engine e, C is the set of components, c is a component of engine e, $t_{compStart}$ is the maintenance start time of the component with the highest priority, P_e is the priority of engine e,$t_R(i)$ is the estimated repair time for engine i.

The swap of an engine is formulated as

$$S_e = \max_{\forall e \in E} R_e + t_s(c) \tag{8}$$

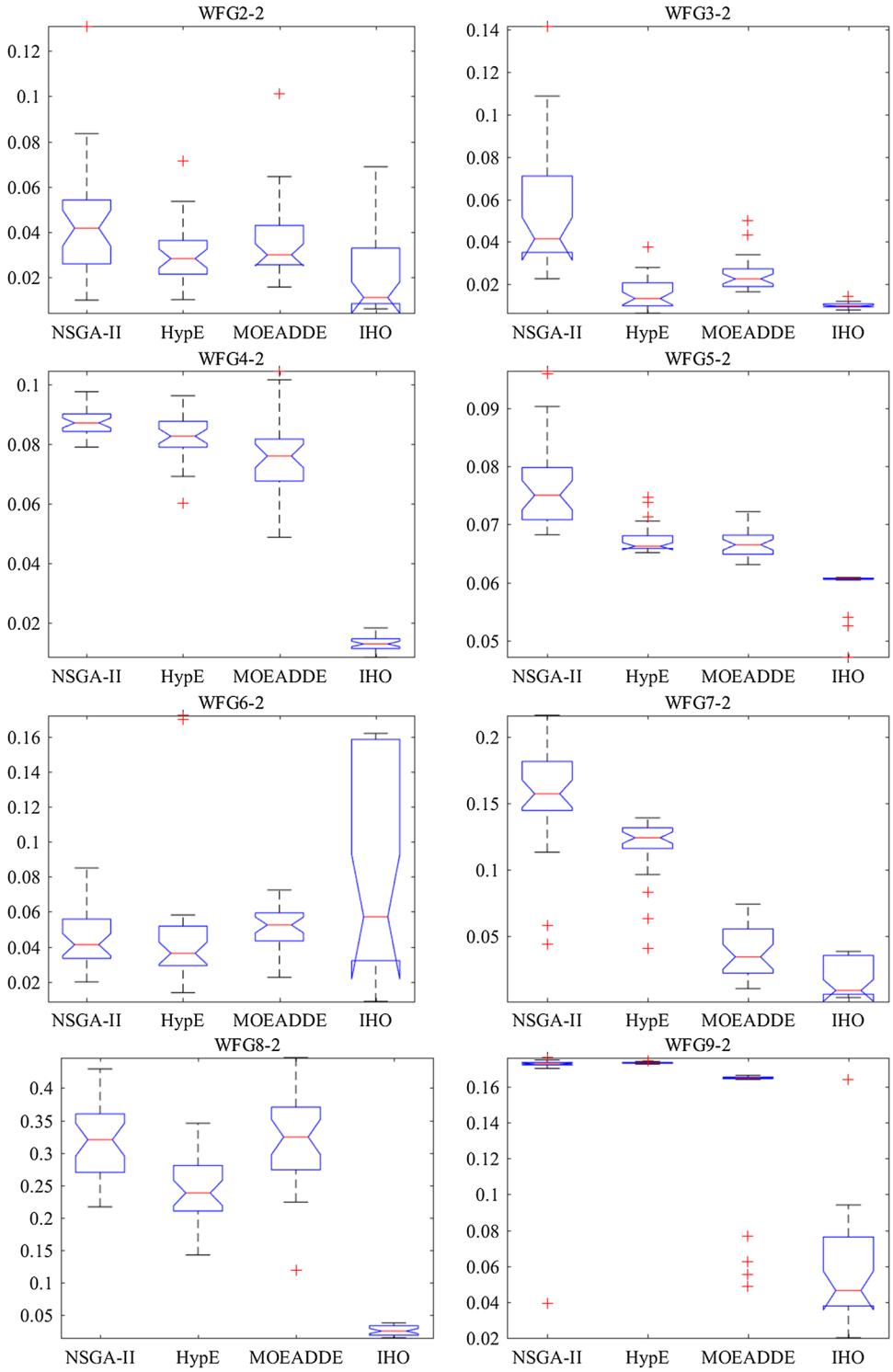


Fig. 4 Box plots of *IGD* results

where S_e is the swap time of engine e, S is the set of 2 engines to be swapped, R_e is the repair time for engine e, $t_s(c)$ is the swap time for component c.

After the swaps, tests are conducted to ensure a proper functionality. The test completion time of each engine is formulated as

$$T_e = S_e + t_{test}(c) \tag{9}$$

where T_e is the test completion time of engine e, S_e is the swap time of engine e, $t_{test}(c)$ is the test time for engine e.

Hence, the first objective function refers to the make-span and is formulated as

$$C_{\max} = \max_{\forall e \in E} T_e \tag{10}$$

where $C_{\max}$ is the makespan, T_e is the test completion time of engine e.

The second objective function refers to the aggregate swap count and is formulated as

$$ASC = \sum_{e=1}^{n} \sum_{k=1}^{5} tol_{ek_{MTTR}} \tag{11}$$

where ASC is the aggregate swap count, n is the total number of engines, $tol_{ek_{MTTR}}$ is the total number of components in engine e that surpasses the disparity threshold of each component.

5.3 Problem settings

The number of engines in this particular problem is set to be 5. Each engine is assigned with arrival time, due time, priority, and mean time between failures (MTBFs) for each of its components. To compute the aforementioned 2 fitness functions, the schedule of each engine and the number of swaps are initialized for optimization. Detailed definition and constraints refer to the reference (Kleeman and Lamont 2007).

The population is set to be 200, and the generation is set to be 200.

5.4 Problem results

As the design variable of MOEA/D-IHO is continuous, here in this particular problem the values are rounded in accordance with the reference (Kleeman and Lamont 2007).

Much of the obtained solutions are duplicate and 7 representative solutions are illustrated in Fig. 5.

Compared to the obtained results in reference (Kleeman and Lamont 2007), the obtained results indicate that the algorithm MOEA/D-IHO is able to obtain more diverse solutions while maintaining a good convergence, which is a valid verification to that conclusion obtained in Sect. 3. With the obtained results, the decision maker is easy to decide which schedule to use.

6 Conclusion and future works

In this paper, we devised a novel MOEA on basis of the MOEA/D framework. In our proposed algorithm, a collaborative indicator-based operator selection is proposed to adaptively select a proper operator to generate the offspring. DE operator and a novel CSO operator are combined to constitute the operator pool. MOEA/D-IHO is

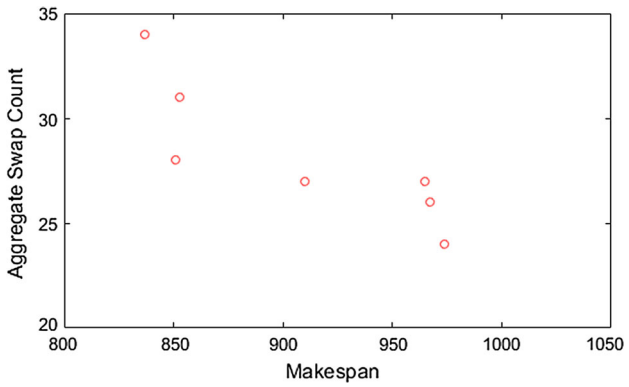


Fig. 5 6 representative obtained non-dominated results

compared with three other elite MOEAs (NSGA-II, HypE, and MOEA/D-DE) on 2 objective WFG2-WFG9 test problems. To eliminate the difference originated by different operator, all these three peer algorithms utilized the DE operator same as our proposed algorithm. Experimental results indicate that MOEA/D-IHO performs almost best in terms of *HV* and *IGD* indicators. In the end, our proposed MOEA/D-IHO is utilized to solve an aeroengine maintenance scheduling problem to indicate the effectiveness of MOEA/D-IHO.

As no further research is conducted to test MOEA/D-IHO in 3 or more objective cases, future research would focus on detailed comparisons with state-of-the-art MOEAs intended for solving MOOPs with more than 3 objectives. Moreover, even though DE is one of the most powerful stochastic optimization algorithms, its performance strongly depends on the proper settings of its parameters. As a result, the fuzzy self-tuning differential evolution (FSTDE) (Tsafarakis et al. 2020), which calculates the settings of parameters independently for each solution during the optimization process, using an automatic FL-based methodology, is strongly suggested to be used in future research. Successful and more recently developed optimization methods like the mayfly optimization algorithm (MA) (Zervoudakis and Tsafarakis 2020) could be used for generating offspring, instead of DE or CSA, in future research.

Abbreviations AOS: Adaptive operator selection; C: Set coverage; PSO: Particle swarm optimization; CSA: Crow search algorithm; CSO: Crow search operator; DE: Differential evolution; HV: Hypervolume; IGD: Inverted generational distance; MOEAs: Multi-objective evolutionary algorithms; MOOP: Multi-objective optimization problem; MaOPs: Many-objective optimization problem; MS: Maximum spread; MTBF: Mean time between failure; GD: Generational distance; PF: Pareto optimal frontier; PM: Polynomial mutation; PS: Pareto optimal set; SBX: Simulated binary crossover

Author contributions GZF performed all experiments and wrote the manuscript, HZH contributed to the conception of the study, YFL helped perform the analysis with constructive discussions, JZ performed the data analyses.

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Compliance with ethical standards

Conflict of interest The authors declare no conflict of interest.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

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